Stroke: Thrombolysis and Endovascular Therapy

Objective:

To outline the evidence, indications and guidance relating to the use of thrombolytic and endovascular therapy for the early management of acute ischemic stroke (AIS).

Background:

Thrombolytic therapy is highly effective in AIS. The recombinant tissue plasminogen activator (rt-PA) alteplase is the most studied systematic agent used in this setting and has demonstrated efficacy in the treatment of AIS in 9 randomized trials involving more than 6,700 patients over more than 20 years. Thrombolytic therapy with intravenous (IV) alteplase has been shown to improve the likelihood of favourable outcome in AIS, despite a small increased risk of serious bleeding. Therapy is time-sensitive, with the benefits of therapy decreasing successively with longer delays from onset to treatment. The benefit is greatest when IV alteplase is administered within the first 3 hours after symptom onset, with a less but still favorable risk-to-benefit ratio for up to 4.5 hours. Treatment benefits of IV thrombolysis are preserved across the spectrum of age, AIS etiology and severity of neurological deficits. The most serious risks of thrombolytic therapy are intracranial hemorrhage (ICH, affecting up to 6% with variable trial-dependent definitions), major extracranial hemorrhage, and angioedema (up to 5%). Approximately 2% of IV thrombolysis-associated intracranial hemorrhages are fatal.

Evidence-based guidelines support the use of thrombolytic therapy with IV alteplase at a dose of 0.9mg/kg for carefully selected patients with disabling AIS syndromes and its use is considered as standard of care in this setting. Five independent randomized trials provided evidence that IV tenecteplase at a dose of 0.25mg/kg has comparable safety and efficacy to alteplase for patients with disabling AIS presenting within 4.5 hours from symptom onset. Given the easier administration scheme of tenecteplase (single IV bolus) many centres in Canada and throughout the world started using tenecteplase as the thrombolytic agent of choice for the treatment of acute ischemic stroke.

Evidence supports the use of advanced neuroimaging to select individuals who may benefit from IV thrombolysis beyond the 4.5-hour window or individuals with unknown time of onset (e.g. those who have disabling deficits without evidence of established parenchymal ischemic changes on MRI, or those having a small established ischemic core and a considerable penumbral brain tissue at risk on CT perfusion imaging). Current Canadian best practice guidelines recommend consultation with a stroke specialist in cases where intravenous thrombolysis is considered to be administered past 4.5 hours from known time of onset.

Endovascular thrombectomy is indicated in selected patients with a proximal intracranial artery occlusion and may be used both in those who have received IV thrombolysis, as well as in those who are not eligible for IV thrombolysis. The benefit exists not only for the treatment of eligible patients within 6 hours from symptoms onset, but also for selected candidates presenting with symptoms up to 24 hours. Randomised clinical trials failed to establish non-inferiority of endovascular treatment alone compared with IV plus endovascular treatment in patients presenting directly at endovascular treatment centres. Therefore, IV thrombolysis with alteplase or tenecteplase should be administered in eligible patients before endovascular treatment.

An effective stroke protocol demands highly coordinated input from paramedics and a pre-notified emergency department team, collaboration from radiology to ensure emergent brain imaging to confirm eligibility, as well as the availability of physicians trained in acute stroke and thrombolytic therapy. If patients may be el

thrombolysis, they should have an emergent head CT at the time of hospital arrival. The target needle time should be less than 30 minutes.

In some regions of the country, a **stroke bypass system** exists in which patients who may have AIS are transported to the nearest stroke center to facilitate rapid expert treatment, with bypass of the closest hospital if they meet established criteria. In most provinces, **Telestroke** is a rapidly developing service to connect remote centers with physicians who have expertise in stroke [https://www.strokebestpractices.ca/recommendations/telestroke].

Patient Selection for Thrombolytic Therapy in AIS:

Inclusion criteria: Patients 18 years or older with a diagnosis of disabling AIS with time of onset ≤4.5 hours. If IV thrombolysis is considered after 4.5 h, consultation with a physician with stroke expertise should be obtained. For adolescents, decision to administer alteplase should be based on clinical judgment, presenting symptoms and patient age. If possible, consultation with a pediatric stroke specialist is recommended.

Absolute Exclusion Criteria:

- Any source of active hemorrhage at a non-compressible site or any condition that could increase the risk of major hemorrhage after thrombolytic administration
- Any intracranial hemorrhage or brain parenchyma neoplasm on imaging

Relative Exclusion Criteria (requiring clinical judgement):

A. History

- History of intracranial hemorrhage
- Ischemic stroke, ST-elevation myocardial infarction, serious head injury or spinal trauma in the preceding 3 months
- Recent major surgery (such as cardiac, thoracic, abdominal, or orthopedic) in the previous 14 days
- Recent gastrointestinal or urinary tract hemorrhage in the previous 21 days
- Arterial puncture at a non-compressible site in the previous 7 days

B. Clinical

- Symptoms suggestive of subarachnoid hemorrhage
- Stroke symptoms due to another non-ischemic acute neurological condition, such as seizure with post-ictal Todd's paralysis or focal neurological signs due to severe hypo- or hyperglycemia (stroke mimics)
- Hypertension refractory to antihypertensive treatment, such that target blood pressure <185/110 cannot be achieved to initiate thrombolysis
- Clincial presentation suggestive of arterial dissection or infective endocarditis
- Currently prescribed and taking a direct oral anticoagulant (DOAC). In centres with access to specialized tests of DOAC levels and reversal agents, thrombolysis could be considered, and decisions made based on individual patient characteristics, in consultation with hematology specialists, patients and their families. Otherwise, direct endovascular treatment should be considered if the patient is otherwise eligible.

C. Laboratory

- Blood glucose concentration below 2.7 mmol/L or above 22.2 mmol/L
- Platelet count <100,000 mm^3

- International Normalized Ratio (INR) > 1.7

D. CT or MRI Findings

- CT or MRI (FLAIR) showing signs of extensive infarction

Thrombolytic Agent Dosing in AIS:

The recommended dose of alteplase for AIS is 0.9 mg/kg (maximum 90 mg) infused over 60 minutes, with 10% of the total dose administered as an initial IV bolus over 1 minute. The recommended dose of tenecteplase is 0.25 mg/kg (maximum 25 mg) administered as a single bolus injection over 5 seconds. Anticoagulants and antiplatelet agents may increase the risk of bleeding and are not recommended within 24 hours of IV alteplase or tenecteplase administration. Any consideration to (re-)start antithrombotic agents should be made after a follow-up brain neuroimaging (CT or MRI) after IV alteplase or tenecteplase administration.

Monitoring of Patients who Received Thrombolytic Therapy for AIS:

The patient should be observed in a monitored setting with frequent neurologic and vital signs assessments as well as cardiac monitoring. Blood pressure should be maintained less than 180/105 mmHg during the administration of IV thrombolysis and for the first 24 hours thereafter. Blood pressure control can be achieved with the use of IV antihypertensive agents. The clinician must be ready to recognize and manage possible complications. Complications include systemic and intracranial bleeding, angioedema and rarely serious anaphylaxis reactions. There is insufficient evidence to support the use of fresh frozen plasma, prothrombin complex concentrates, or platelets for rt-PA related bleeding; however, cryoprecipitate and tranexamic acid are recommended in addition to supportive care. The decision to administer these agents should be balanced against the risk for ischemic events.

Endovascular Therapy:

Endovascular therapy (EVT) may be used in eligible patients alone or in combination with IV thrombolysis. There is very strong evidence from multiple randomized trials demonstrating benefit of EVT in individuals with proximal intracranial occlusions either in the anterior circulation (internal carotid artery and proximal middle cerebral artery as well as individuals with tandem cervical carotid and intracranial occlusions) or the posterior circulation (vertebral artery and/or basilar artery) who present with moderate-to-severe neurological deficits. The majority of patients from pivotal EVT trials were treated within 6 hours of stroke onset, and shorter times from onset to recanalization are associated with better functional outcomes. Most participants in these trials also received IV alteplase. More recent trials have now established the benefit of EVT in individuals treated up to 24 hours from time of last seen well and those found to have large infarction on neuroimaging. Two independent trials have recently suggested no benefit and increased risks with EVT for patients presenting with occlusions in smaller intracranial vessels. The benefit of EVT in individuals with milder neurological deficits and proximal large artery occlusions is being evaluated in ongoing clinical trials.

Treatment decision-making regarding suitability for EVT is generally a collaborative effort between the neuro-interventionalist and stroke physician. Decision-making takes into account the patient’s pre-morbid status, current neurological deficits, burden of ischemic core on neuroimaging, and vascular access. Transport logistics are also often germane to treatment decisions as EVT is performed only at comprehensive stroke centres. Success of EVT requires the coordination of pre-hospital and hospital systems, rapid neurovascular imaging, expertise in neurointervention, and access to critical care units (step down units, intensive care units).

Special Considerations:

Pediatrics: There are little safety and efficacy data in patients under 18 years of age with AIS. expertise in pediatric stroke should manage these patients where possible. When this is not possible, a combination of a pediatrician and an adult neurologist, supported by consultation with an experienced specialist in pediatric stroke, is recommended.

Pregnancy: There have been no empiric studies of IV thrombolysis in pregnant women presenting with AIS. However, it is reasonable to consider administering IV thrombolysis to a pregnant woman with disabling AIS who meets existing criteria for treatment. Pregnancy should not be considered to be a contraindication for EVT. Although there are fetal risks to these interventions, given the very high morbidity and mortality associated with AIS, maternal care for disabling stroke should not be delayed or deferred on the basis of pregnancy. Treatment options for pregnant women with AIS should promptly be considered in consultation with an interdisciplinary team with expertise in neurology, obstetrics and gynecology, maternal-fetal medicine, and interventional radiology where possible and available.

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https://www.strokebestpractices.ca/recommendations/acute-stroke-management/acute-ischemic-stroke-treatment

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